

- Models would frequently call the end task tool very quickly.

The final system message we used for BasicAgent is displayed in Figure 10. The agent is fed the task instructions (see Appendix F.3) via a user message.

F.2. IterativeAgent

Despite our improvements made to the original scaffold when developing BasicAgent, we found that most models used with BasicAgent still intentionally used the submit tool to end the task early. Interestingly, most of the time models justified this choice by claiming that they were instructed to complete a partial reproduction of the paper, rather than a full reproduction.

We developed IterativeAgent from BasicAgent to encourage models to complete the entire task. Every time we queried the model we instructed it to only take the next step towards replicating the paper; we found this to significantly reduce the likelihood of models finishing early. We also removed the submit tool so IterativeAgent would have to work for the full time available.

IterativeAgent uses a different system message displayed in Figure 11. If the model produces a message with no tool calls we append the user message in Figure 12 to the message history.

F.3. Task Instructions

We report the task instructions for our benchmark in Figure 13 and Figure 14. We make no rule about how the task instructions should be ingested by a submitting agent scaffold (e.g., scaffolds may choose to ignore this particular formulation of the instructions and prompt the backend model their own way), and provide them as part of our benchmark.

G. More on JudgeEval

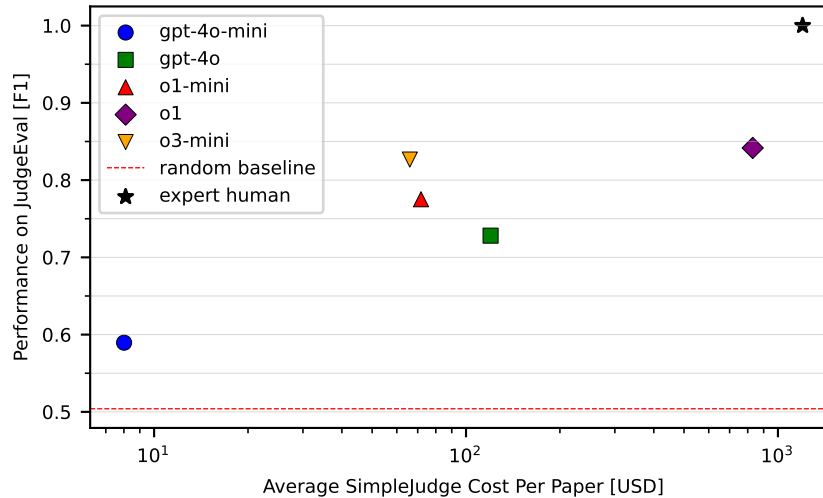


Figure 5. Performance on JudgeEval vs average cost per paper in JudgeEval for various model backends in SimpleJudge. Model cost measured in terms of input+output tokens multiplied by their respective cost-per-token on the OpenAI API. Human cost estimated at 12 hours of work at an hourly rate of \$100 USD/hr. Reasoning models are run with reasoning effort set to “high”.

In Fig. 5, we plot the average performance (F1) vs. cost (\$USD per paper) for various model backends when using SimpleJudge on JudgeEval. We estimate model cost in terms of API usage cost (number of input/output tokens multiplied by the relevant cost-per-token based on public OpenAI o1 API pricing as of 2025/03/21)

We also plot the expert human performance and cost, treating this as ideal performance and estimating cost at 12 hours of work per paper at a hypothetical hourly rate of 100 \$USD/hr. Finally, we plot the performance of a random baseline where the judge randomly marks leaf nodes as satisfied or unsatisfied.

We find that humans are hundreds of dollars more costly than the most expensive model (o1) for end users. Additionally, we

find performance comparable to o1 with o3-mini, at one-tenth of the cost.

Table 8. Macro-averaged F1-score for the models evaluated as part of JudgeEval with the SimpleJudge scaffold. We report F1 both overall and stratified across requirement types.

	OVERALL	CODE DEVELOPMENT	EXECUTION	RESULT MATCH
RANDOM BASELINE	0.49	0.45	0.48	0.46
SIMPLEJUDGE				
GPT-4O-MINI	0.59	0.59	0.54	0.78
GPT-4O	0.73	0.68	0.70	0.83
O1-MINI-HIGH	0.77	0.67	0.74	0.80
O1-HIGH	0.84	0.74	0.84	0.88
O3-MINI-HIGH	0.83	0.72	0.82	0.94

In Table 8 we accompany the overall F1 scores reported in Table 3 with their stratified counterparts, measuring how well SimpleJudge performs based on requirement type.

We see that performance is relatively stable across requirement types, although a clear gradient of “difficulty” is also discernible: models struggle most on Code Development nodes and perform best on Result Match nodes. O3-MINI-HIGH achieves an F1-score of 0.72 in Code Development, which we deem acceptable for tracking signal on this requirement type. We note that O1-HIGH seems to outperform O3-MINI-HIGH on Code Development and Execution, although this difference may be due to noise and remains futile given the much higher costs.

H. Pruned Rubric Grading

As pointed out in Section 7 and in Appendix G, grading PaperBench rollouts can be quite expensive. Despite o3-mini’s reduced token costs, grading still costs around \$66 per paper on average. While PaperBench Code-Dev offers an alternative, it does come at the cost of foregoing the Execution and Result Match tasks.

As an alternative approach to reduce the cost and time necessary for grading, we experimented with “pruning” the rubrics shown to the judge, collapsing trees past a certain depth into a single leaf node. To do the collapsing, past a certain depth we simply concatenate the contents of the subtrees and subleafs of the current node.

This greatly reduces the amount of output tokens required from the Judge, as the number of decisions is reduced: rather than reasoning and grading all leaf nodes with a binary score, past a certain depth the Judge simply grades entire sub-trees, assigning a float score between 0 and 1. This, however, makes the grading task more difficult for the judge as it must assess in one go whether multiple requirements have been achieved.

In Figure 6, we show the overall replication score assigned by the Judge to one of the submissions in our JudgeEval (see Section 4.2) when pruning at different depths. We observe that, for this paper and submission, pruning anything beyond depth 3 already approaches the score that would be achieved in the default case with no pruning. We note that pruning at depth 3 reduced grading cost by 10×, while the performance of the Judge deteriorates only slightly.

While this is promising, we wish to underline that these are preliminary results on an experimental version of the Judge, and we have also observed cases of unsatisfactory performance. We expect that as models get better, we can increasingly move towards grading subtrees as opposed to grading leaves.

I. Full Results

In Tables 10, 11, 12, 13, 14, 15, 16, 17, and 18, we display each of the agent performances on each of the 20 papers in our dataset. Notably, for most agents, we see high variance in results on the same paper. Due to the high variance, we recommend others to use several seeds when evaluating PaperBench to get an accurate measure of agent performance.

I.1. Results stratified by requirement type

See Table 9 for results stratified by requirement type. We observe that models perform poorly on Execution and Result Match requirement types, while scoring better at Code Development nodes. This suggests that models are good at writing

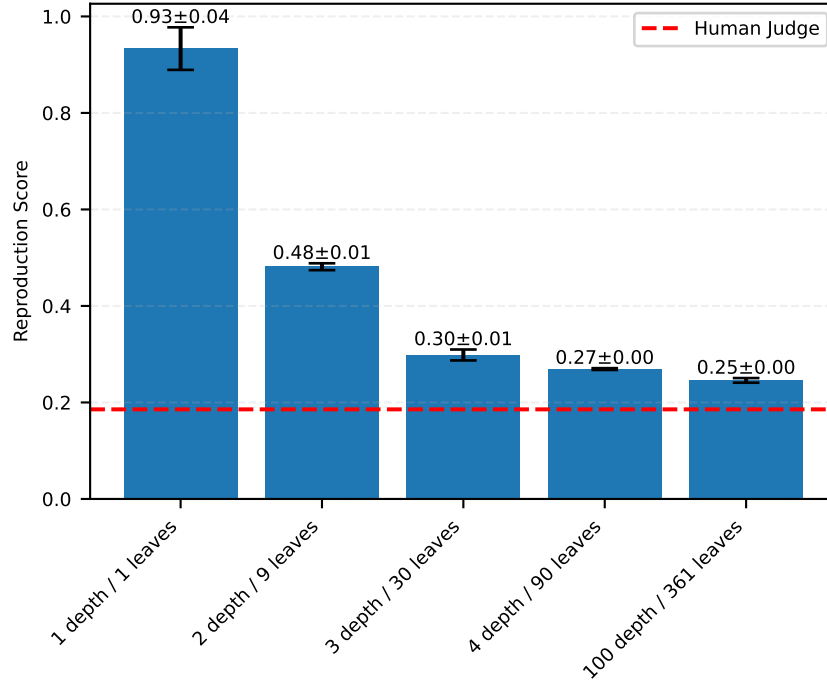


Figure 6. The replication score assigned by o3-mini SimpleJudge to the ‘rice/0’ submission in JudgeEval, over different depths of pruning. Pruning at depth 100 is equivalent to not pruning for this paper. We plot the ground truth human Judge grade in red. Error bars are standard error of the mean over 3 repeats.

lots of code, but aren’t successful at integrating, testing, and successfully running that code to achieve results.

Table 9. Average Replication Scores for models with our scaffold for each requirement type. Standard error is computed across three seeds, except for o1 (IterativeAgent) and gemini-2.0-flash (BasicAgent) where it is computed across two seeds.

MODEL	CODE DEVELOPMENT	EXECUTION	RESULTS ANALYSIS
GEMINI-2.0-FLASH (BASICAGENT)	5.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
4O (BASICAGENT)	7.7 ± 0.0	0.1 ± 0.1	0.0 ± 0.0
O3-MINI (BASICAGENT)	5.1 ± 0.8	0.6 ± 0.1	0.4 ± 0.4
O1 (BASICAGENT)	19.5 ± 1.2	5.7 ± 0.9	0.0 ± 0.0
R1 (BASICAGENT)	9.8 ± 0.0	1.0 ± 0.8	0.0 ± 0.0
CLAUDE-3-5-SONNET (BASICAGENT)	35.4 ± 0.8	1.8 ± 0.7	0.7 ± 0.3
O3-MINI (ITERATIVEAGENT)	16.4 ± 1.4	0.6 ± 0.4	0.3 ± 0.1
O1 (ITERATIVEAGENT)	43.3 ± 1.1	4.5 ± 1.5	0.0 ± 0.0
CLAUDE-3-5-SONNET (ITERATIVEAGENT)	27.5 ± 1.6	1.1 ± 0.1	0.9 ± 0.4
O1 [36 HOURS] (ITERATIVEAGENT)	42.4 ± 1.0	7.4 ± 1.1	1.4 ± 0.1
BEST@3 HUMAN [3 PAPER SUBSET]	72.4	20.4	8.9

Table 10. GPT-4o BasicAgent results. * indicates a result that was set to 0% due to disqualification violating PaperBench rules.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0*	0.027	0.193	0.073	0.049
ALL-IN-ONE	0.008	0.025	0.015	0.016	0.004
BAM	0.131	0.000	0.089	0.074	0.032
BBOX	0.002	0.018	0.002	0.007	0.004
BRIDGING-DATA-GAPS	0.003	0.011	0.004	0.006	0.002
FRE	0.029	0.014	0.016	0.020	0.004
FTRL	0.070	0.021	0.000	0.030	0.017
LBCS	0.081	0.033	0.000	0.038	0.019
LCA-ON-THE-LINE	0.000	0.013	0.051	0.021	0.013
MECHANISTIC-UNDERSTANDING	0.019	0.056	0.053	0.042	0.010
PINN	0.112	0.058	0.000	0.057	0.026
RICE	0.000	0.000	0.015	0.005	0.004
ROBUST-CLIP	0.233	0.178	0.193	0.201	0.014
SAMPLE-SPECIFIC-MASKS	0.000	0.192	0.118	0.103	0.046
SAPG	0.031	0.000	0.000	0.010	0.009
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.048	0.000	0.000	0.016	0.013
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.034	0.084	0.026	0.048	0.015
STOCHASTIC-INTERPOLANTS	0.020	0.049	0.005	0.024	0.011
TEST-TIME-MODEL-ADAPTATION	0.005	0.028	0.002	0.012	0.007
WHAT-WILL-MY-MODEL-FORGET	0.051	0.000	0.000	0.017	0.014

Table 11. o1 BasicAgent results.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.037	0.107	0.037	0.060	0.019
ALL-IN-ONE	0.098	0.091	0.041	0.077	0.014
BAM	0.255	0.206	0.297	0.253	0.021
BBOX	0.117	0.109	0.139	0.122	0.007
BRIDGING-DATA-GAPS	0.135	0.098	0.136	0.123	0.010
FRE	0.155	0.000	0.064	0.073	0.037
FTRL	0.014	0.000	0.036	0.017	0.008
LBCS	0.322	0.166	0.275	0.254	0.038
LCA-ON-THE-LINE	0.024	0.062	0.073	0.053	0.012
MECHANISTIC-UNDERSTANDING	0.000	0.132	0.231	0.121	0.055
PINN	0.239	0.129	0.078	0.149	0.039
RICE	0.245	0.099	0.078	0.141	0.043
ROBUST-CLIP	0.145	0.126	0.128	0.133	0.005
SAMPLE-SPECIFIC-MASKS	0.448	0.229	0.098	0.258	0.083
SAPG	0.101	0.083	0.102	0.095	0.005
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.076	0.304	0.318	0.233	0.064
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.060	0.057	0.056	0.058	0.001
STOCHASTIC-INTERPOLANTS	0.326	0.182	0.230	0.246	0.035
TEST-TIME-MODEL-ADAPTATION	0.094	0.069	0.176	0.113	0.027
WHAT-WILL-MY-MODEL-FORGET	0.107	0.041	0.060	0.069	0.016

Table 12. o1 IterativeAgent results.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.249	0.140	0.238	0.209	0.028
ALL-IN-ONE	0.220	0.043	0.182	0.148	0.044
BAM	0.296	0.464	0.387	0.383	0.040
BBOX	0.201	0.175	0.212	0.196	0.009
BRIDGING-DATA-GAPS	0.218	0.189	0.189	0.199	0.008
FRE	0.290	0.373	0.276	0.313	0.025
FTRL	0.093	0.106	0.096	0.098	0.003
LBCS	0.463	0.464	0.209	0.379	0.069
LCA-ON-THE-LINE	0.182	0.144	0.138	0.155	0.011
MECHANISTIC-UNDERSTANDING	0.272	0.440	0.197	0.303	0.059
PINN	0.214	0.000	0.289	0.168	0.071
RICE	0.152	0.207	0.070	0.143	0.032
ROBUST-CLIP	0.202	0.183	0.161	0.182	0.010
SAMPLE-SPECIFIC-MASKS	0.425	0.465	0.466	0.452	0.011
SAPG	0.188	0.193	0.148	0.177	0.012
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.544	0.408	0.447	0.466	0.033
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.132	0.176	0.109	0.139	0.016
STOCHASTIC-INTERPOLANTS	0.397	0.319	0.266	0.327	0.031
TEST-TIME-MODEL-ADAPTATION	0.220	0.237	0.245	0.234	0.006
WHAT-WILL-MY-MODEL-FORGET	0.308	0.151	0.175	0.212	0.040

Table 13. o3-mini BasicAgent results.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.012	0.012	0.007	0.010	0.001
ALL-IN-ONE	0.000	0.022	0.062	0.028	0.015
BAM	0.059	0.042	0.029	0.043	0.007
BBOX	0.004	0.012	0.018	0.011	0.003
BRIDGING-DATA-GAPS	0.011	0.000	0.010	0.007	0.003
FRE	0.005	0.010	0.000	0.005	0.002
FTRL	0.000	0.000	0.008	0.003	0.002
LBCS	0.057	0.029	0.040	0.042	0.007
LCA-ON-THE-LINE	0.031	0.060	0.038	0.043	0.007
MECHANISTIC-UNDERSTANDING	0.028	0.028	0.028	0.028	0.000
PINN	0.078	0.060	0.148	0.095	0.022
RICE	0.039	0.004	0.000	0.014	0.010
ROBUST-CLIP	0.000	0.000	0.032	0.011	0.009
SAMPLE-SPECIFIC-MASKS	0.123	0.000	0.000	0.041	0.033
SAPG	0.000	0.009	0.003	0.004	0.002
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.000	0.000	0.059	0.020	0.016
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.010	0.178	0.039	0.075	0.042
STOCHASTIC-INTERPOLANTS	0.005	0.078	0.029	0.037	0.017
TEST-TIME-MODEL-ADAPTATION	0.008	0.002	0.005	0.005	0.001
WHAT-WILL-MY-MODEL-FORGET	0.010	0.000	0.001	0.003	0.003

Table 14. o3-mini IterativeAgent results.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.054	0.076	0.063	0.064	0.005
ALL-IN-ONE	0.030	0.036	0.090	0.052	0.016
BAM	0.087	0.100	0.082	0.089	0.004
BBOX	0.086	0.060	0.009	0.051	0.018
BRIDGING-DATA-GAPS	0.023	0.015	0.012	0.017	0.003
FRE	0.066	0.050	0.020	0.045	0.011
FTRL	0.059	0.017	0.011	0.029	0.012
LBCS	0.077	0.076	0.079	0.077	0.001
LCA-ON-THE-LINE	0.100	0.053	0.013	0.055	0.021
MECHANISTIC-UNDERSTANDING	0.093	0.064	0.075	0.077	0.007
PINN	0.125	0.109	0.135	0.123	0.006
RICE	0.025	0.005	0.002	0.011	0.006
ROBUST-CLIP	0.149	0.037	0.119	0.102	0.027
SAMPLE-SPECIFIC-MASKS	0.168	0.167	0.231	0.189	0.017
SAPG	0.089	0.056	0.035	0.060	0.013
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.680	0.542	0.144	0.455	0.131
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.050	0.062	0.037	0.050	0.006
STOCHASTIC-INTERPOLANTS	0.084	0.020	0.156	0.087	0.032
TEST-TIME-MODEL-ADAPTATION	0.036	0.091	0.059	0.062	0.013
WHAT-WILL-MY-MODEL-FORGET	0.000	0.000	0.038	0.013	0.010

Table 15. Claude 3.5 Sonnet BasicAgent results.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.133	0.188	0.176	0.166	0.014
ALL-IN-ONE	0.267	0.284	0.194	0.248	0.023
BAM	0.199	0.371	0.187	0.252	0.049
BBOX	0.135	0.185	0.000	0.107	0.045
BRIDGING-DATA-GAPS	0.173	0.203	0.099	0.158	0.025
FRE	0.141	0.265	0.273	0.227	0.035
FTRL	0.114	0.091	0.073	0.093	0.010
LBCS	0.128	0.103	0.364	0.198	0.068
LCA-ON-THE-LINE	0.134	0.189	0.095	0.140	0.022
MECHANISTIC-UNDERSTANDING	0.075	0.222	0.245	0.181	0.043
PINN	0.183	0.329	0.231	0.248	0.035
RICE	0.202	0.231	0.163	0.198	0.016
ROBUST-CLIP	0.273	0.301	0.315	0.296	0.010
SAMPLE-SPECIFIC-MASKS	0.246	0.369	0.314	0.309	0.029
SAPG	0.040	0.036	0.087	0.054	0.013
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.365	0.459	0.420	0.414	0.022
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.106	0.085	0.135	0.108	0.012
STOCHASTIC-INTERPOLANTS	0.155	0.071	0.147	0.124	0.022
TEST-TIME-MODEL-ADAPTATION	0.182	0.105	0.143	0.143	0.018
WHAT-WILL-MY-MODEL-FORGET	0.382	0.256	0.253	0.297	0.035

Table 16. Claude 3.5 Sonnet IterativeAgent results.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.214	0.292	0.165	0.224	0.030
ALL-IN-ONE	0.163	0.115	0.066	0.115	0.023
BAM	0.223	0.237	0.101	0.187	0.035
BBOX	0.157	0.128	0.170	0.152	0.010
BRIDGING-DATA-GAPS	0.057	0.086	0.100	0.081	0.010
FRE	0.167	0.158	0.200	0.175	0.010
FTRL	0.042	0.041	0.044	0.043	0.001
LBCS	0.300	0.232	0.089	0.207	0.051
LCA-ON-THE-LINE	0.170	0.108	0.122	0.133	0.015
MECHANISTIC-UNDERSTANDING	0.041	0.014	0.056	0.037	0.010
PINN	0.113	0.399	0.163	0.225	0.072
RICE	0.073	0.114	0.047	0.078	0.016
ROBUST-CLIP	0.254	0.220	0.268	0.247	0.012
SAMPLE-SPECIFIC-MASKS	0.326	0.225	0.221	0.257	0.028
SAPG	0.110	0.053	0.024	0.063	0.021
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.433	0.388	0.159	0.327	0.069
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.089	0.077	0.319	0.162	0.064
STOCHASTIC-INTERPOLANTS	0.192	0.278	0.320	0.263	0.031
TEST-TIME-MODEL-ADAPTATION	0.144	0.130	0.151	0.142	0.005
WHAT-WILL-MY-MODEL-FORGET	0.156	0.089	0.098	0.114	0.017

Table 17. Gemini 2.0 Flash BasicAgent results. Run 3 of what-will-my-model-forget failed due to infrastructure issues. * indicates a result that was set to 0% due to disqualification violating PaperBench rules.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.036	0.060	0.173	0.090	0.034
ALL-IN-ONE	0.000	0.007	0.089	0.032	0.023
BAM	0.000	0.000	0.323	0.108	0.088
BBOX	0.000	0.000	0*	0.00	0.00
BRIDGING-DATA-GAPS	0.011	0.013	0.025	0.016	0.004
FRE	0*	0*	0.007	0.061	0.009
FTRL	0.047	0*	0*	0.016	0.013
LBCS	0.114	0.039	0.092	0.082	0.018
LCA-ON-THE-LINE	0.000	0.038	0.010	0.016	0.009
MECHANISTIC-UNDERSTANDING	0.052	0.000	0.021	0.024	0.012
PINN	0.034	0.012	0.000	0.015	0.008
RICE	0.013	0.000	0.007	0.007	0.003
ROBUST-CLIP	0.000	0.132	0.214	0.115	0.051
SAMPLE-SPECIFIC-MASKS	0.069	0.122	0.006	0.066	0.027
SAPG	0.015	0.010	0.024	0.016	0.003
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.160	0.222	0.284	0.222	0.029
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.000	0.000	0.036	0.012	0.010
STOCHASTIC-INTERPOLANTS	0.000	0.000	0.059	0.020	0.016
TEST-TIME-MODEL-ADAPTATION	0*	0.000	0.000	0.000	0.000
WHAT-WILL-MY-MODEL-FORGET	0.030	0.056	–	0.043	0.009

Table 18. R1 BasicAgent results. * indicates a result that was set to 0% due to disqualification violating PaperBench rules.

PAPER	RUN 1	RUN 2	RUN 3	MEAN	STD. ERROR
ADAPTIVE-PRUNING	0.133	0.040	0.046	0.073	0.025
ALL-IN-ONE	0.065	0.049	0.139	0.085	0.023
BAM	0.087	0.017	0.123	0.075	0.025
BBOX	0.048	0.058	0.023	0.043	0.008
BRIDGING-DATA-GAPS	0.027	0.055	0.000	0.027	0.013
FRE	0.071	0.020	0*	0.030	0.017
FTRL	0.000	0*	0.019	0.006	0.005
LBCS	0.036	0.025	0.000	0.020	0.009
LCA-ON-THE-LINE	0.000	0.003	0.015	0.006	0.004
MECHANISTIC-UNDERSTANDING	0.052	0.000	0.000	0.017	0.014
PINN	0.008	0.000	0.101	0.036	0.026
RICE	0*	0.000	0.025	0.008	0.007
ROBUST-CLIP	0.114	0.056	0.000	0.057	0.027
SAMPLE-SPECIFIC-MASKS	0.110	0.201	0.050	0.120	0.036
SAPG	0.030	0.000	0.032	0.021	0.009
SEQUENTIAL-NEURAL-SCORE-ESTIMATION	0.327	0.212	0.552	0.364	0.082
STAY-ON-TOPIC-WITH-CLASSIFIER-FREE-GUIDANCE	0.073	0.053	0.083	0.070	0.007
STOCHASTIC-INTERPOLANTS	0.072	0.263	0.068	0.134	0.053
TEST-TIME-MODEL-ADAPTATION	0.000	0.006	0.022	0.009	0.005
WHAT-WILL-MY-MODEL-FORGET	0.000	0.000	0.005	0.002	0.001